

Syllabus for LAMP-L316

Instructor: Robert Granger (rgranger@iu.edu)

Office Hours: see [Office Hours](#).

Communication Policy: Send me an email.

Textbook: No textbook, but there will be readings from select open-source textbooks and websites.

Required Materials: Class is in a computer lab, but it may be beneficial to have a laptop with a strong internet connection.

Required Software: We will be using R with Rstudio. I will show you how to install this on your own personal machines, or you can use it on the computers provided in lab.

Important Dates:

- Automatic Withdrawal Deadline: Sunday, October 25, 2026
- Initial Meeting with Clients: October 12-16, 2026 (TBA)
- Final Presentations: December 14-18, 2026 (TBA)

Course Objectives:

Description: Analytical Problem Solving is a 300-level, service-learning course that trains LAMP students to use quantitative methods and analytic skills to explore and solve business-related problems. Through classroom instruction, weekly labs, and a project for a real-world client, students develop their statistical skills, learn to be savvy consumers of statistics, and learn to assess and interpret statistical results.

CO1: Organize, summarize, and visualize data using descriptive statistics, tables, and graphs to identify meaningful patterns and relationships.

CO2: Conduct and interpret statistical inference, including confidence intervals and hypothesis tests, while clearly explaining the assumptions, limitations, and practical significance of the results.

CO3: Critically evaluate statistical claims and analyses presented in business, academic, and public contexts, including the quality of the data, the appropriateness of the methods, and the strength of the conclusions.

CO4: Collaborate with a real-world client to define an analytical problem, analyze relevant data, and communicate evidence-based recommendations through clear written, oral, and visual presentations.

Note on Course Content: The focus of this course is learning statistical methods. Within that focus, I have selected class materials that present a variety of topics involving

business, politics, natural sciences, and many more. It is possible that connections between our course and current and/or historical events outside the classroom will come up in lectures, class discussions, and assignments. The ability to draw and analyze connections among ideas and events is a core skill in a liberal arts education and will stand you in good stead in your life beyond IU.

Course Components and Grading

Attendance/Participation: 20%

Attendance will be recorded in various ways at every class meeting and will be awarded only when you are present and actively engaged for the entire class period. If your level of engagement is insufficient, your instructor will notify you.

If you must miss class for a valid reason, contact your instructor as soon as possible, preferably before class.

In-Class Lab Activities: 40%

On most Fridays, as indicated on the course calendar, you will complete an in-class activity. There will be ten activities in total, and each must be completed during the designated class period. Your lowest activity score will be dropped when calculating your final activity grade.

Client Presentation: 40%

Working with a real-world client, you will analyze data, prepare a written report, develop a presentation, and present your findings and recommendations to the client. Your grade will reflect the quality of your work across all components of the project.

Grading Scale:

A+ 98% or above
A from < 98% to 92%
A- from < 92% to 90%
B+ from < 90% to 88%
B from < 88% to 82%
B- from < 82% to 80%
C+ from < 80% to 78%
C from < 78% to 72%
C- from < 72% to 70%
D+ from < 70% to 68%
D from < 68% to 62%
D- from < 62% to 60%
F below 60%

Accommodations

If you are in need of an accommodation, it is your duty to inform your instructor. We would appreciate it very much if you ***inform us during the first two weeks of the term***. If a disability or illness arises during the term, you should inform your instructor at your earliest possible convenience.

Course Accessibility: Indiana University is dedicated to ensuring that students with disabilities (e.g., chronic health, neurodevelopmental, neurological, sensory, psychological, and emotional, including mental health, etc.) have the support services and reasonable accessibility measures (accommodations) needed to provide equal access to academic programs. You must establish your eligibility for support services through [Accessible Educational Services \(AES\)Links to an external site.](#). Note that services are confidential, may take time to put into place, and are not retroactive; citations and alternate media for print materials may take three or more weeks to produce. Contact Accessible Educational Services as soon as possible to [register for servicesLinks to an external site.](#). The office is located on the basement floor of Eigenmann Hall in Room 001 and is open Monday through Friday, 8 a.m. to 5 p.m.

Religious Observation:

Instructors must reasonably accommodate students who want to observe their religious holidays at times when academic requirements conflict with those observances. If you will miss class for a religious observance, you must fill out the [Request for Accommodation for Religious Observances formLinks to an external site.](#) at least two weeks before your anticipated absence.

Pregnancy: IU is dedicated to supporting students who are pregnant or have experienced other conditions related to pregnancy (termination of pregnancy, miscarriage, lactation, or related medical conditions). To register for accommodations due to pregnancy or a related medical condition, please contact [Accessible Educational ServicesLinks to an external site.](#)

Major Absences: If you miss an exam or a substantial portion of classes, you must contact your instructor as soon as possible. Your absence needs to be documented. In some cases, such as hospitalization or a death in the family, the [Dean of StudentsLinks to an external site.](#) will help verify the documentation. Accommodations may involve nothing (so you get a zero on the exam if you miss the exam), re-weighting the assignments that were not missed, allowing a make-up assignment, or another accommodation.

Incomplete Policy: A student may receive an "I" (Incomplete) grade in exceptional cases where valid reasons hinder the completion of the course. After successful completion of the missing assignments, the letter grade will be updated to reflect the grade earned in the course. Per [IU policyLinks to an external site.](#), students have one calendar year from the date an Incomplete is recorded to resolve it, unless the dean of their college or school grants an extension under exceptional circumstances. The registrar will automatically convert the "I" to an "F" after the designated time period unless an extension has been approved or the student has earned a degree during that time. The

student is responsible for reaching out to their instructor to complete the course. If the instructor is unavailable, the student should attempt to contact the Director of Business Statistics or the Statistics Department Chair.

Academic Integrity:

You should take some time to review the [Code of Student Rights, Responsibilities, and Conduct \(Code\)](#)[Links to an external site.](#) because you are expected to adhere to its standards while you are a student at IU. Holding students to the standards outlined in the Code ensures the value of the degree that you are earning from IU and is important training for maintaining ethical standards in the work that you will do in the future.

Academic misconduct is “any activity that tends to undermine the academic integrity of the institution” (Code, II, B,1). Academic integrity violations include cheating, fabrication, plagiarism, interference, violation of course rules, facilitating academic dishonesty, and research misconduct. When you submit an assignment with your name on it, you are signifying that the work contained therein is yours, unless otherwise cited or referenced. Any ideas or materials taken from another source must be fully acknowledged. Do not share your completed work with other students. If plagiarism or cheating occurs, all students involved will be considered responsible even if you were unaware that academic misconduct would occur. Ignorance of what constitutes academic misconduct is not a valid excuse. All suspected violations of the Code will be reported to the Dean of Students (Office of Student Conduct). Sanctions for academic misconduct in this course may include a failing grade on the assignment, a reduction in your final course grade, or a failing grade in the course, among other possibilities. If you are unsure about the expectations for completing an assignment or taking a test or exam in this course, speak with me before submitting your assignment.

Students who commit academic misconduct in any IU class will not be able to utilize the grade replacement policy on those enrollments. Grades in courses in which students have committed academic misconduct will always be factored into their GPAs.

Note Selling: Various commercial services have approached students regarding selling class notes/study guides to their classmates. Selling the instructor’s notes/study guides or uploading course assignments to these sites (even after the course has ended) in exchange for access to materials for other courses is not permitted. Violations of this policy will be reported to the Dean of Students (Office of Student Conduct) as academic misconduct. Sanctions for academic misconduct for this action may include a failing grade on the assignment for which the notes/study guides or assignments have been being uploaded, a reduction in your final course grade, or a failing grade in the course, among other possibilities. Additionally, you should know that selling a faculty member’s notes/study guides individually or on behalf of one of these services using IU email or via Canvas, may also constitute a violation of IU information technology and IU intellectual property policies; additional consequences may result.

Online Course Materials: The instructor teaching this course holds the exclusive right to distribute, modify, post, and reproduce course materials, including all written materials, study guides, lectures, assignments, exercises, and exams. Some of the course content may be downloadable, but you should not distribute, post, or alter the instructor's intellectual property. While you are permitted to take notes on the online materials and lectures posted for this course for your personal use, you are not permitted to re-post in another forum, distribute, or reproduce content from this course without the express written permission of the instructor.

GroupMe and Other External Messaging Platforms: Please note that you may receive emails from other students about joining GroupMe, Discord, or similar external group messaging platforms for individual classes via Canvas. Even though invitations to join the group may be issued through Canvas, they do not imply the endorsement of the course instructor. While platforms like GroupMe, Discord, etc. can be an effective way of contacting classmates and clarifying information related to the course, they can also be source of unauthorized information sharing or collaboration among students. Collaborative effort on assignments, quizzes, and exams, including sharing or discussing answers when the instructor has not expressly authorized collaboration, is considered cheating. If academic dishonesty occurs via GroupMe or a similar messaging platform, everyone involved in the thread may be found responsible for academic misconduct since membership in the group suggests that they have been able to view the information shared.

Other University Information

FERPA

The [Family Educational Rights and Privacy Act \(FERPA\)](#)Links to an external site. is a federal law that protects the privacy of your education records. You may authorize a third party to access your education record by adding a release at <https://ferpareleases.indiana.edu>Links to an external site.. Adding a release for a specific individual will generate a PIN that is sent to the student's IU email and will direct the student to share the PIN with the authorized third party.

Health and Wellbeing

- The [Student Care and Resource Center](#)Links to an external site. supports students experiencing challenges that impact their ability to be safe, healthy, and successful. You can walk in, call, email, submit a [Care Referral](#)Links to an external site. or make an appointment through the Student Appointment Scheduler to connect with the Care Team.
- Mental Health Support: If you or someone you know needs assistance, you are encouraged to contact Counseling and Psychological Services (CAPS) at 812-855-5711 or by [visiting their webpage](#)Links to an external site.. They will

help you navigate any difficult circumstances you may be facing by connecting you with the appropriate resources or services.

Bias-Based Incident Reporting:

As your instructor, one of my responsibilities is to create a positive learning environment for all students. Bias incidents (events or comments that target an individual or group based on age, color, religion, disability, race, ethnicity, national origin, sex, gender, gender identity, sexual orientation, marital status or veteran status) are not appropriate in our classroom or on campus. What should you do if you witness or experience a bias incident? See it? Hear it? Report it by submitting a report online (biasincident.indiana.edu ([Links to an external site.](#))) or calling the Dean of Students Office (812-855-8187).

Sexual Misconduct and Title IX:

IU policy prohibits sexual misconduct in any form, including sexual harassment, sexual assault, stalking, sexual exploitation, and dating and domestic violence. If you have experienced sexual misconduct, or know someone who has, the University can help. If you are seeking help and would like to speak to someone confidentially, you can make an appointment with the IU Sexual Assault Crisis Services at (812) 855-8900 or contact a Confidential Victim Advocate at (812) 856-2469 or cva@iu.edu.

It is also important that you know that University policy requires that I share certain information brought to my attention about potential sexual misconduct with the campus Deputy Sexual Misconduct & Title IX Coordinator or the University Sexual Misconduct & Title IX Coordinator. In that event, those individuals will work to ensure that appropriate measures are taken, and resources are made available. Protecting student privacy is of the utmost concern, and information will only be shared with those that need to know to ensure the University can respond and assist. I encourage you to visit <http://stopsexualviolence.iu.edu/index.html> ([Links to an external site.](#)) to learn more.

Student Resources:

[Library Research Assistance](#) ([Links to an external site.](#)): IU Libraries offers one-on-one consultations with students who need assistance with library research.

[Student Advocates](#) ([Links to an external site.](#)): Retired IU faculty and staff members assist students with resolving academic, financial, and conduct-related issues.

Technology-related resources provided by the university include:

- Free on-campus wireless internet (wifi) access through the “eduroam” network
- Free software for download and for cloud-based use (access the Software Downloads app in [One.IU](#) ([Links to an external site.](#)))
- Free 24/7 IU tech support (e.g., email, Canvas, wifi, printing, device setup, etc.)
 - Chat: <https://ithelplive.iu.edu/> ([Links to an external site.](#))
 - Phone: 812-855-6789

- Email: ithelp@iu.edu
- Web form: [Submit an IT support requestLinks to an external site.](#)
- Online: [Knowledge BaseLinks to an external site.](#)

- Free in-person tech support in the Wells Library

[Tutoring ResourcesLinks to an external site.](#): There are many free and paid tutoring resources on campus that are available to students.

[Writing Tutorial ServicesLinks to an external site.](#): Offers free one-on-one writing help from a trained tutor.